

8 Lab 7: Setup Optimization with VI-CRT

8.1 Introduction

This laboratory has the aim of understanding the methodologies for race car setup optimization using the VI-grade software, with a specific focus on the **VI-CarRealTime (VI-CRT)** environment. VI-CRT provides an advanced simulation platform that allows engineers to model vehicle subsystems, perform multi-body dynamics analyses, and predict lap time performance under various operating conditions and setups.

The primary objective of this laboratory is to explore the setup of a **Dallara F312**, analyzing its performance on a specific racetrack. By iterating through different setup configurations, the laboratory aims to demonstrate the complexity of vehicle optimization, how parameters influence each other and the necessary trade-offs.

8.2 Track Layout and Path Creation

The first step before starting the simulations is setting the racetrack environment.

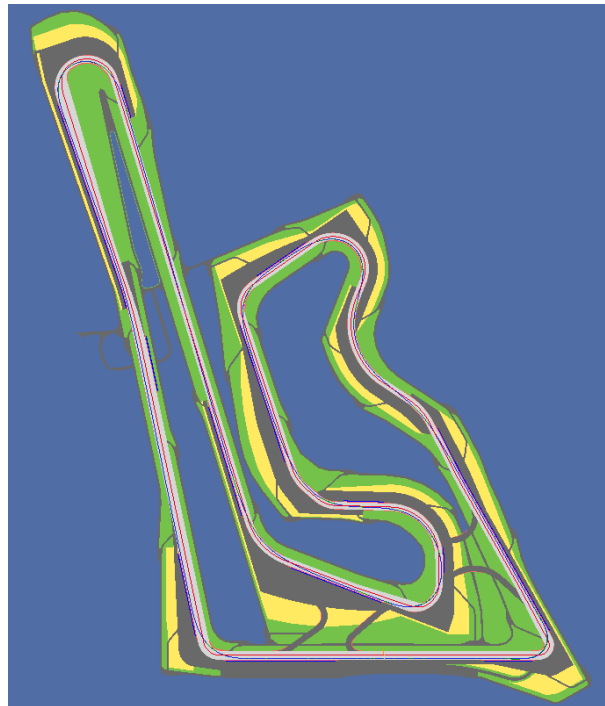


Figure 127: Track Layout

The vehicle's trajectory depends on the approach we want the driver to follow during the lap simulation; generally, a **Corner Cutting Algorithm** is implemented to optimize the overall path and travel the shortest possible distance

The algorithm considers a trajectory point valid if its distance from the centerline satisfies:

$$D \leq C_{cut} - \frac{W - V}{2}$$

where W is the road width, V is the vehicle width, and C_{cut} is the corner cutting value.

The path is then generated by optimizing a cost function that balances path tension and curvature: $Cost = \int (\alpha \cdot PathTension^2 + \beta \cdot PathCurvature^2) ds$. By adjusting the tension and curvature weights (β and α), the trajectory can be made shorter or smoother depending on the desired driver line.

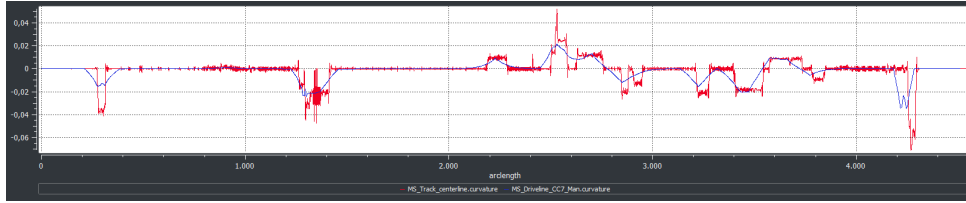
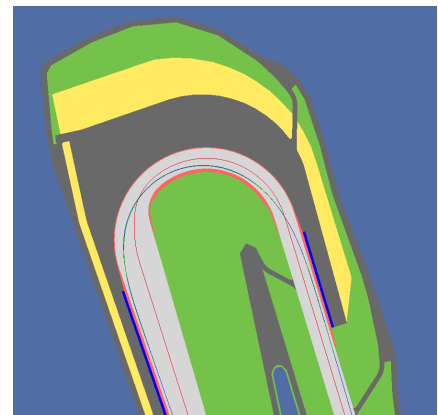
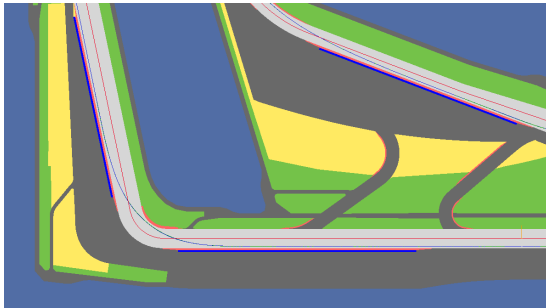


Figure 128: Center-Line vs Corner-Cut Trajectories



Finally, a **Smoothing Algorithm** is applied to remove noise on both lateral curvature and track height variation to ensure a clean path for the simulation. The smoothing technique is essential to clean every kind of numerical fluctuation, which can significantly alter the model performance and responses along the racetrack.

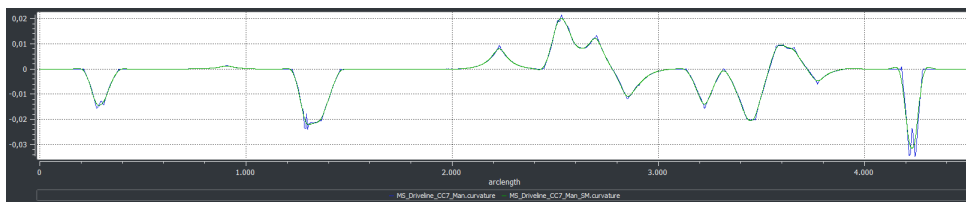


Figure 130: Curvature: Baseline vs Smoothed Trajectory

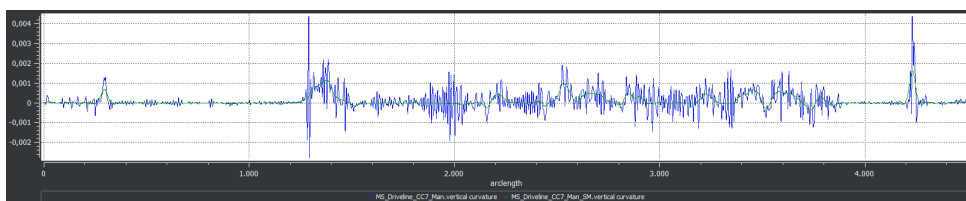


Figure 131: Track Height Variation: Baseline vs Smoothed Trajectory

8.3 SpeedGen vs MaxPerformance Event

Two main event types are used to compute lap times: SpeedGen and MaxPerformance events.

- **SpeedGen (Static Lap Time):** This event computes a quasi-static simulation using a simplified vehicle model which does not account for complex dynamic scenarios.

The vehicle follows the given path and computes the maximum speed profile according to grip limit. If the path profile is not smoothed, the vehicle might follow any kind of fluctuation present, reducing the overall performance.

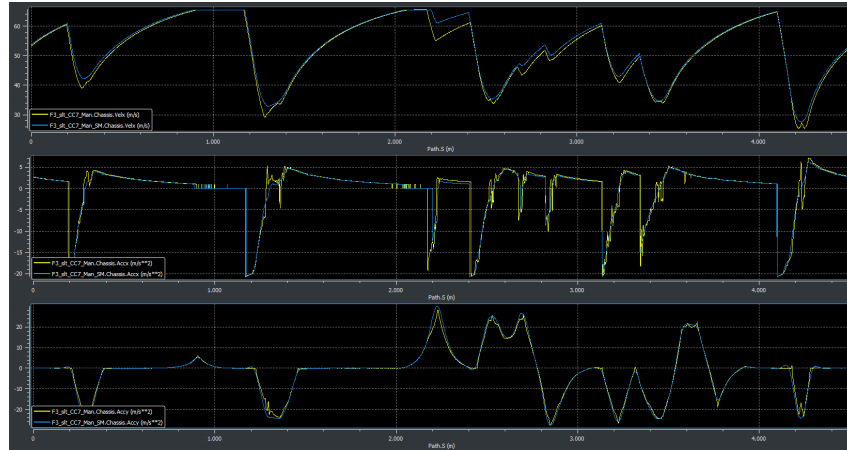


Figure 132: QSS simulation following a baseline trajectory bs smoothed one

Path	Predicted LapTime [s]
Baseline Corner Cut	90.282485
Smoothed Corner Cut	88.082701

- **MaxPerformance (Dynamic Lap Time):** This event performs a fully dynamic simulation. It uses the SpeedGen profile as a target. The simulation is performed iteratively by dividing the overall race track into sectors. If the dynamic simulation leads to an unfeasible situation, the algorithm restores the vehicle state until a valid, feasible lap is completed.

Moreover, a target acceleration in lateral and longitudinal direction can be given to challenge the model in finding a more competitive solution. This increases the iteration process whenever the vehicle is not able to sustain such accelerations.

Sector	0	1	2	3	4	5	6	7	8	9	Total
N° iter (1.1g)	0	3	0	2	0	0	1	0	0	1	7
N° iter (1.2g)	5	8	0	7	0	3	3	0	0	4	30

Table 25: 1.1g vs 1.2g Acceleration Target

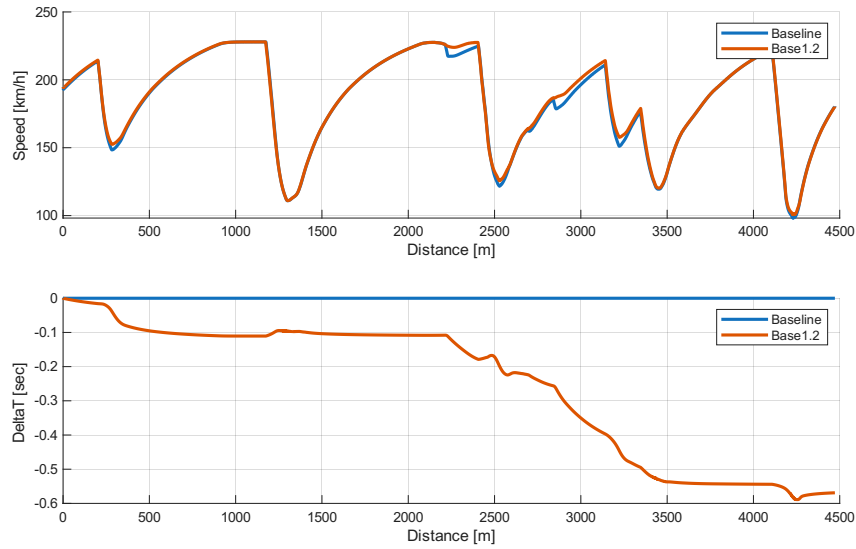


Figure 133: 1.1g vs 1.2g Acceleration Target

Setup	Elapsed LapTime [s]
Baseline1.1	90.261
Baseline1.2	89.691

8.4 Setup Comparisons

The vehicle overall setup depends on many parameters coming from different subsystems. The reciprocal influence makes the setup choice really challenging since it usually leads to trade-offs and tuning optimization.

8.4.1 Long vs Short Gear Setup

From the Baseline Setup, it can be noticed from the speed profile that the vehicle is not exploiting its maximum speed; this behavior is related to the gear ratio set adopted. In the Baseline case, a short gear ratio set helps the car to be more responsive on the throttle in traction, however, it limits the top speed because the limiter is reached too early.

To solve this, a longer gear ratio set is adopted to exploit the maximum top speed and avoid reaching the limiter

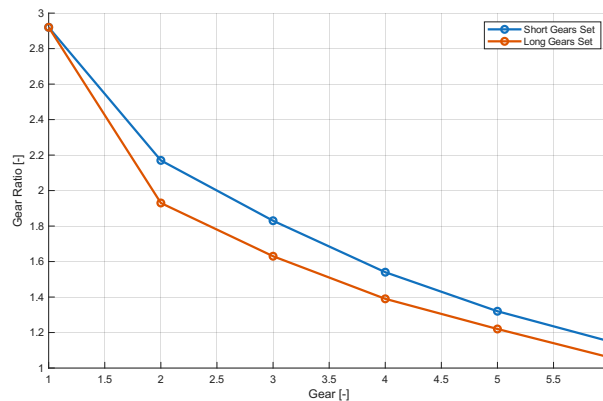


Figure 134: Long vs Short Gear Ratio Set

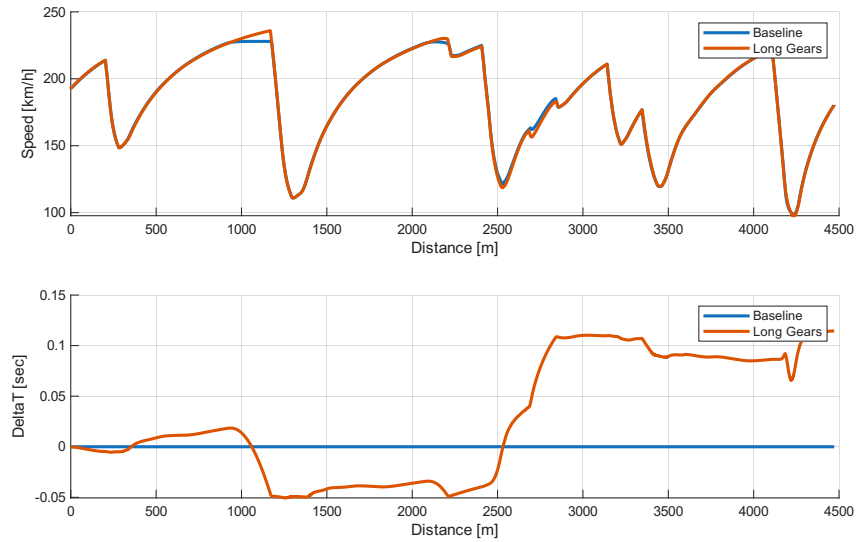


Figure 135: Short vs Long Gear setup

Setup	Elapsed LapTime [s]
Short Gear Set	90.261
Long Gear Set	90.371

It's interesting to see that even if the Long Gear Setup is able to avoid hitting the rev limiter and exploit the top speed on the main straight (useful for defense and overtaking), the Short Gear Setup is still faster at the end of the lap due to its responsiveness in the slowest section of the circuit, which allows the car to have a better torque at lower speed (so in acceleration phases, especially out of corners).

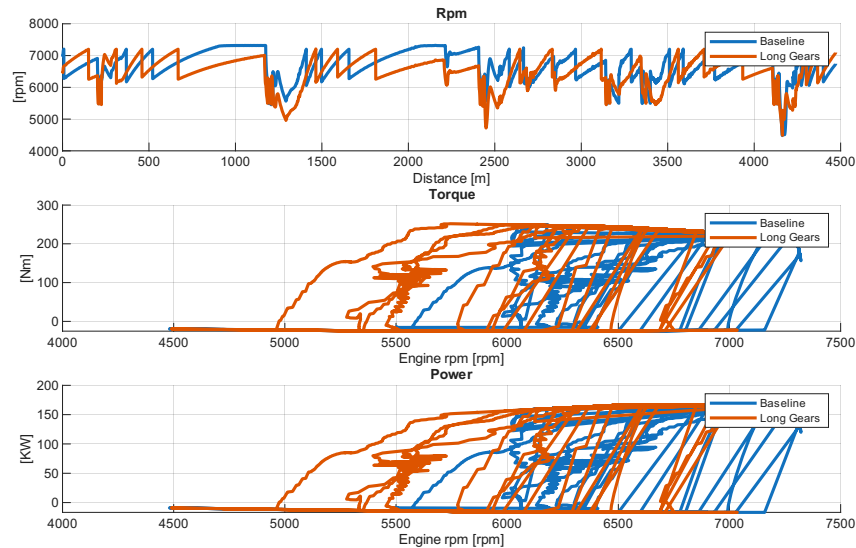


Figure 136: Engine Plots

8.4.2 High vs Low Downforce Setup

Two different Aero Maps are compared, a Low Downforce Setup allows the vehicle to have higher top speed on the straights, while a High Downforce Setup is more efficient in

corners.

Configuration	High Downforce (HD)			Low Downforce (LD)		
RRH [mm]	Drag [N]	F. DF [N]	R. DF [N]	Drag [N]	F. DF [N]	R. DF [N]
5	1439	1714	2406	1295	1457	2093
15	1455	1818	2439	1310	1545	2122
25	1470	1876	2427	1323	1595	2111

Table 26: Aero Forces (at 180 km/h), fixed Front RH (10mm), variable Rear RH.

HD generates much more overall downforce at the cost of increased drag, while maintaining a slight advantage in overall aerodynamic efficiency.

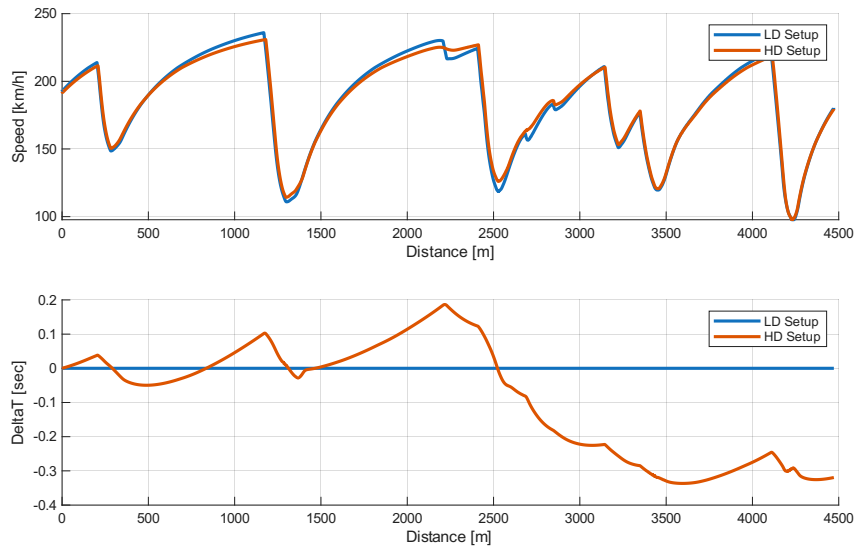


Figure 137: LD vs HD Setup

Setup	Elapsed LapTime [s]
Low Downforce	90.371
High Downforce	90.061

Although the LD setup allows the vehicle to be faster in the first part of the circuit, as soon as the track becomes more technical, the HD setup is able to take curves at a significantly higher speed, delay the braking point by a few meters and have a better minimum speed at the center of the curve.

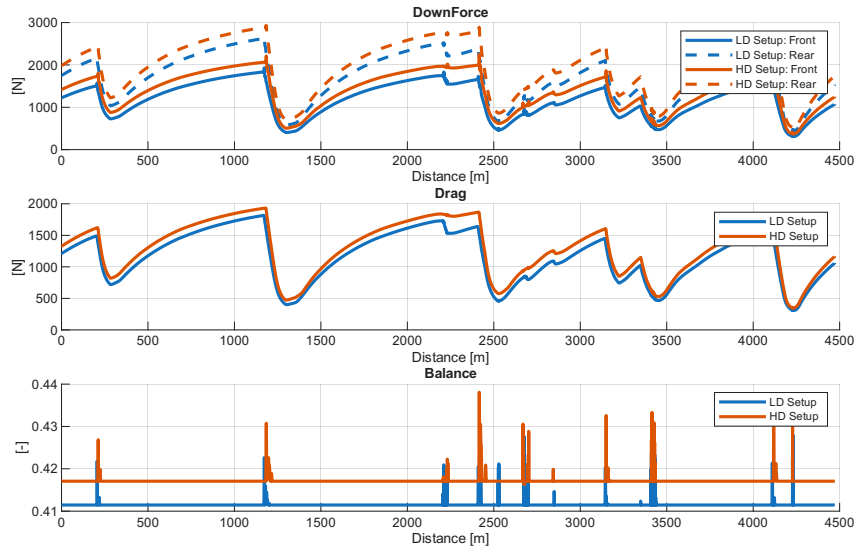


Figure 138: Aero Plots

8.4.3 RARB Adjustment: Under vs Over-steering tendency

Two different Anti-Roll Bars are mounted on the Rear Axle. The first one (ScalingFactor of 1.0) is less rigid while the second one (ScalingFactor 2.0) is stiffer. A first evident effect is on the rear-end roll angle in corners.

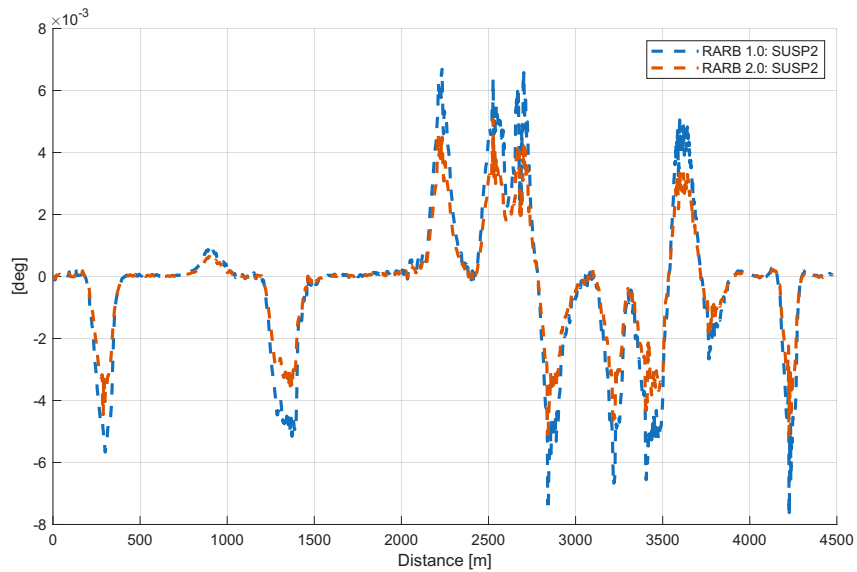


Figure 139: Rear Suspension Roll Angle

The Anti-Roll bar has a primary effect on the axle load transfer capacity and, as a consequence, on the overall vehicle behavior.

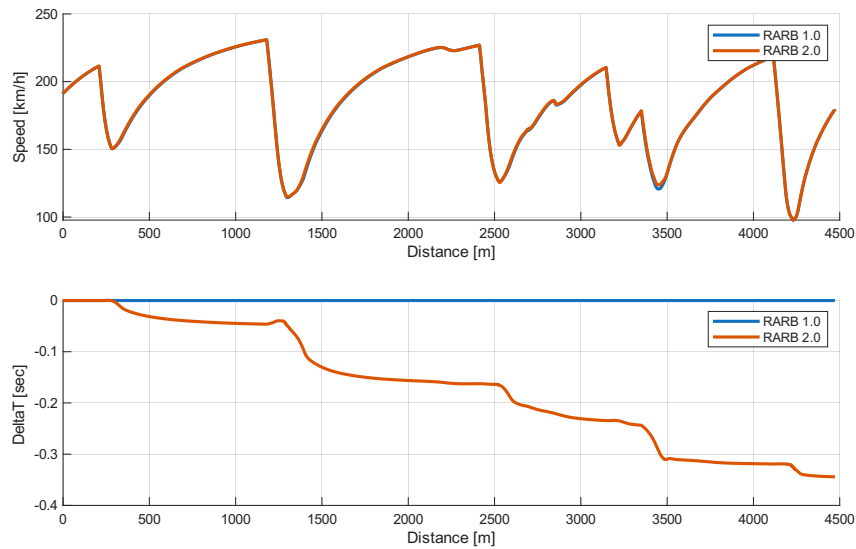


Figure 140: RARB 1.0 vs RARB 2.0 Setup

Setup	Elapsed LapTime [s]
RARB 1.0	90.061
RARB 2.0	89.711

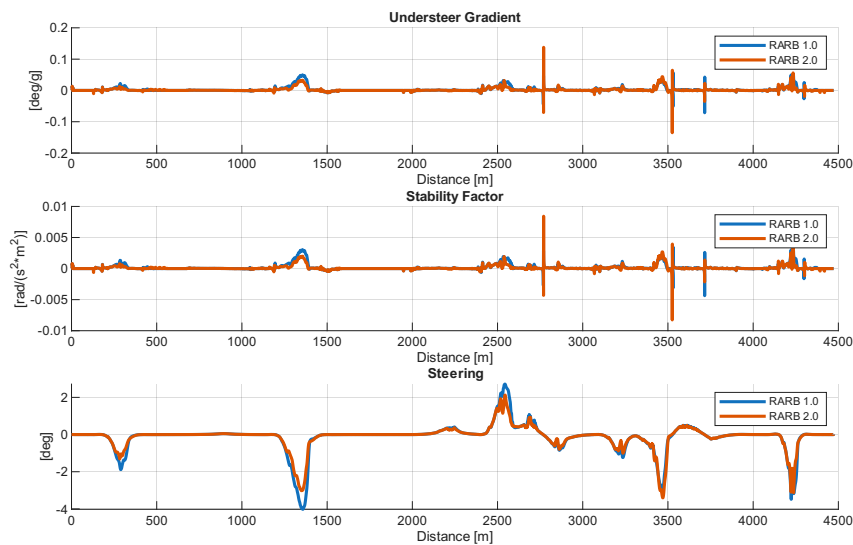


Figure 141: Vehicle Stability Plot

Even if the difference cannot be noticed from the pure speed telemetry, the Anti-Roll Bar adjustment has a huge impact on the driver sensibility and driving style. A Stiffer Rear Anti-Roll Bar makes the Rear Axle overall cornering capability less efficient. The car becomes more Over-steering, which of course is bad for stability but at the same time helps the vehicle to be more reactive in direction changes. The driver needs less steering angle to perform corners, especially in the slow ones, where a reactive car is needed to turn as quickly as possible to go earlier on the throttle.

8.5 Design Of Experiment Analysis (DOE)

To optimize the vehicle setup, a Design of Experiments (DOE) analysis is conducted using the Investigation Mode within VI-CarRealTime. The DOE approach replaces the previous trial-and-error tuning process with a structured mathematical methodology to evaluate how multiple variables interact and affect the vehicle's overall performance, primarily focusing on the elapsed lap time.

For this laboratory, a **Full Factorial** experimental design is implemented. In a full factorial design, the software evaluates every possible combination of the selected parameters. For each investigated parameter, three discrete levels were defined: a lower limit, an intermediate baseline value, and an upper limit. This matrix allows for the evaluation of the vehicle responses to parameter variations, identifying the optimal setup region within the assigned boundaries.

8.5.1 Toe Angle

The following DOE analysis is conducted to evaluate the effects of static Toe angles on the front and rear axles. Toe settings primarily govern the vehicle's directional stability, turn-in responsiveness, and straight-line rolling resistance.

From a theoretical vehicle dynamics perspective, the setup choice for front and rear toe angles are as follows:

- **Front Axle:** A slight **Toe-out** ($\delta_{front} < 0$) is generally preferred. This geometry pre-loads the inner tire with a favorable slip angle during the initial steering input, significantly enhancing turn-in response and mitigating entry understeer.
- **Rear Axle:** A slight **Toe-in** ($\delta_{rear} > 0$) is crucial. Rear toe-in acts as a stabilizing mechanism, generating a restoring yaw moment that keeps the rear stable during heavy braking and aggressive throttle application at corner exits, preventing snap oversteer.

3 levels of toe angle are applied to the Front and Rear wheels.

Level	δ_{front} [°]	δ_{rear} [°]
Low	-0.18	0.123
Medium	-0.13	0.173
High	-0.08	0.223

The DOE investigation provides the following results:

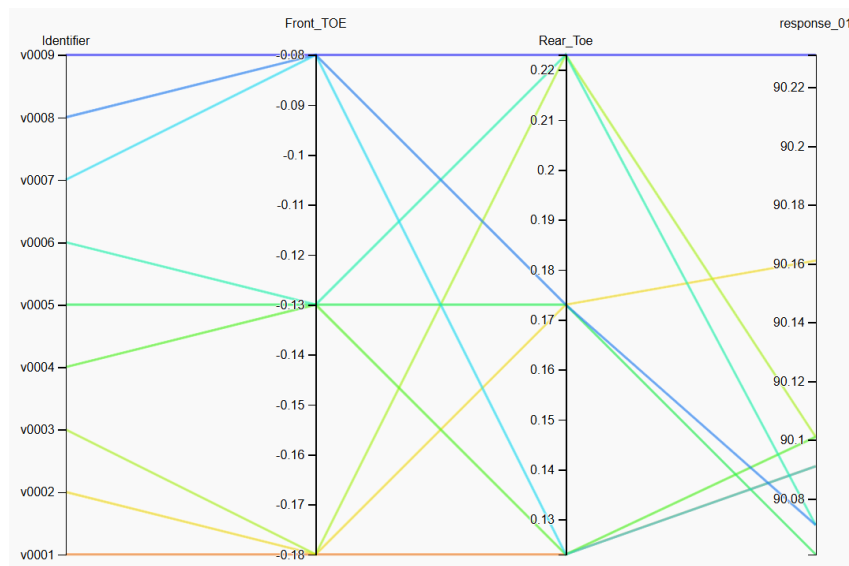


Figure 142: Toe DOE combinations

v00x	δ_{front} [°]	δ_{rear} [°]	Elapsed LapTime [s]
v0005 (Baseline - Best)	-0.13	+0.173	90.061

The simulation reveals that the Baseline setup (v0005) remains the optimal configuration, representing the best compromise between cornering grip, stability, and rolling resistance.

To better understand the vehicle dynamics behind this result, a telemetry comparison was performed between the optimal baseline (v0005) and configuration v0001. Configuration v0001 features a more aggressive front Toe-out (-0.180°) and a reduced rear Toe-in (0.123°).

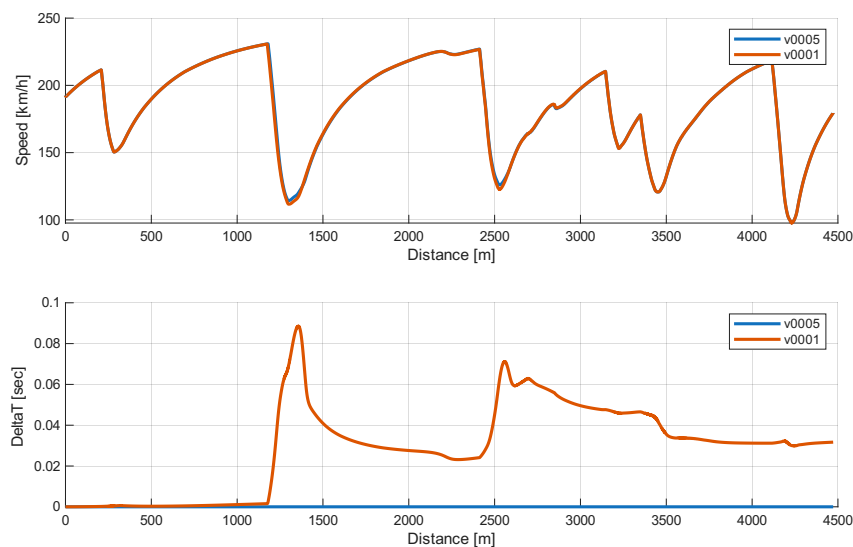


Figure 143: v0005 (Baseline) vs v0001 Setup

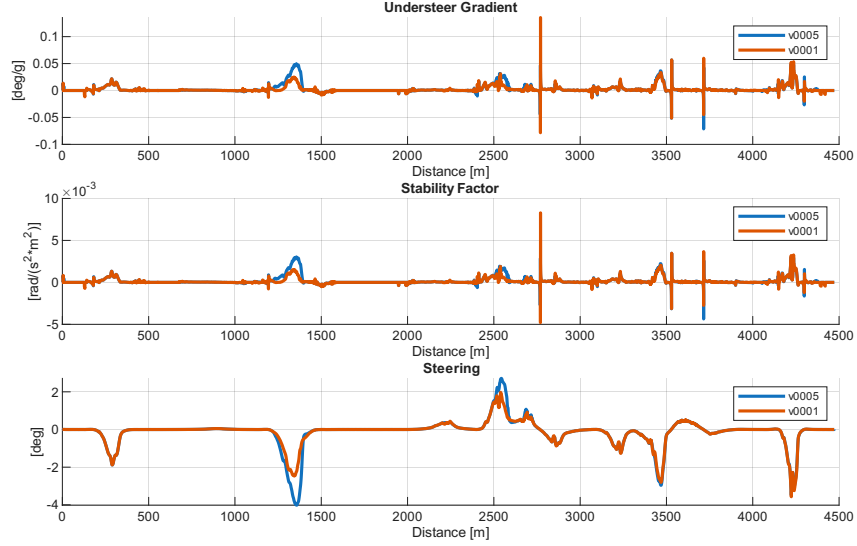


Figure 144: Vehicle Stability Plot

v00x	δ_{front} [°]	δ_{rear} [°]	Elapsed LapTime [s]
v0005 (Baseline - Best)	-0.13	+0.173	90.061
v0001	-0.18	+0.123	90.091

The telemetry highlight two distinct behaviors:

- **Heavy Braking (v0001 Time Loss):** The v0001 setup is significantly slower during heavy braking zones. The reduction in rear toe-in compromises the stabilizing yaw moment during deceleration, forcing the virtual driver to brake earlier and less aggressively to prevent rear-end instability.
- **Slow Corners Entering (v0001 Time Gain):** The v0001 setup gains time in corner entry due to a more responsive front-end: a slightly larger toe-out helps the directionality on the vehicle and reduce the steering effort required by the driver.
- **Traction and Straights (v0001 Time Gain):** The v0001 setup gains time during pure longitudinal acceleration and on long straights. By reducing the rear toe-in, the rear tires are aligned straighter relative to the direction of travel and the transmission of torque on the ground is more efficient

8.5.2 Stiffness Optimization and Ride Heights Check

A final investigation is performed to find an optimal solution which is able to extract the maximum Lap Time performance without touching the ground (Bottoming).

To solve this constrained optimization problem, 4-factor, 3-level full factorial DOE is configured. The variables included are:

1. Front Suspension Stiffness (K_{front}) Scaling Factor
2. Rear Suspension Stiffness (K_{rear}) Scaling Factor
3. Front Anti-Roll Bar (FARB) Stiffness Scaling Factor

4. Rear Anti-Roll Bar (RARB) Stiffness Scaling Factor

Each parameter is evaluated at a Lower (0.7), Medium (1.0), and Upper (1.3) limit, generating a total of 81 unique setup combinations.

The optimization targets the minimization of the elapsed Lap Time while imposing a strict physical constraint: the dynamic ground clearance. To prevent the vehicle from bottoming, three height sensors mounted on the car are monitored:

- Front Left RH Sensor
- Front Right RH Sensor
- Rear Center RH Sensor

The evaluation of the 81 configurations generated by the combined DOE brings highly significant results regarding the vehicle's mechanical balance. The data analysis focuses on three key configurations: the Baseline (v0041), the Best Infeasible Setup (v0009), and the Best Legal Setup (v0026).

Configuration	LapTime [s]	FARB	K_{front}	RARB	K_{rear}	Min RH Registered [m]	Status
Best Infeasible (v0009)	89.571	0.7	0.7	1.3	1.3	-0.0052	Invalid (Bottoming)
Best Legal (v0026)	89.911	0.7	1.3	1.3	1.0	+0.0002	Valid
Baseline (v0041)	90.061	1.0	1.0	1.0	1.0	-0.0008	Invalid (Bottoming)

Table 27: DOE Results

A surprising finding from this investigation is that the original **Baseline setup (v0041)** is physically infeasible, recording negative dynamic ride heights. This highlights the absolute necessity of the optimization process.

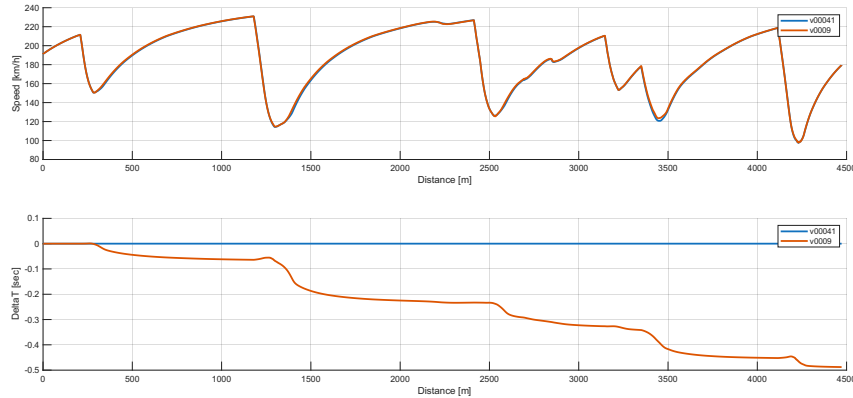


Figure 145: v0041 (Baseline) vs v0009 (Best Infeasible) Setup

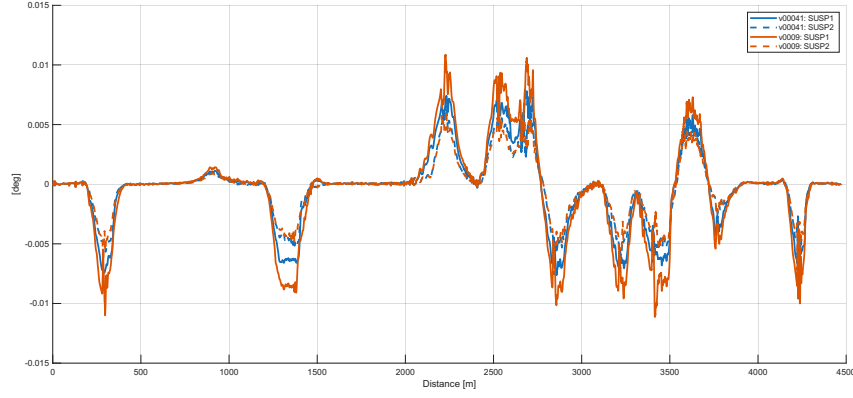


Figure 146: Suspension Roll Plot

The **Theoretical Fastest Setup** (v0009) adopts extreme softness at the front (K_{front} and FARB at 0.7 of Scaling Factor) combined with maximum stiffness at the rear (K_{rear} and RARB at 1.3). This specific combination shifts the roll stiffness distribution to the rear axle.

The front-end gains in mechanical grip while the rear-end is more rigid, resulting in an oversteering and reactive car, which is crucial for performance extraction.

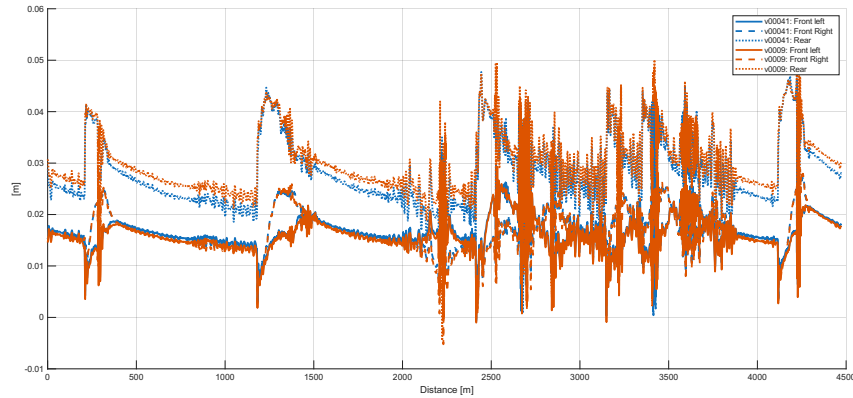


Figure 147: Ride Heights Plot

However, the extremely soft front suspension allows excessive dive under aerodynamic load and braking pitch. Consequently, the front crashes into the ground (minimum front RH of -5.2 mm), making this mathematically optimal setup physically invalid and undriveable in reality.

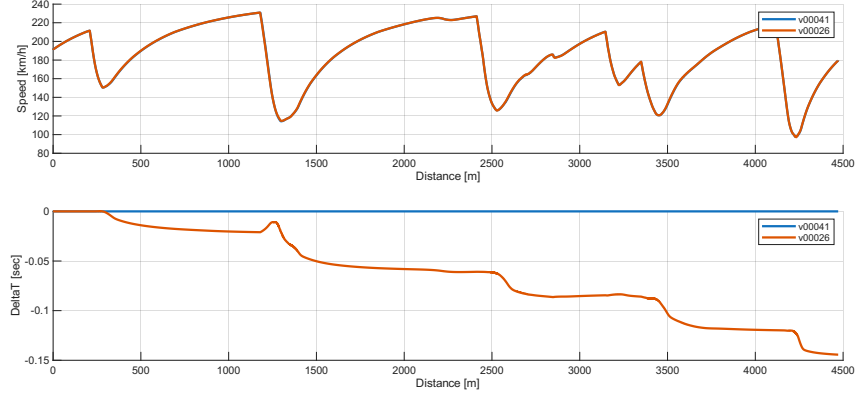


Figure 148: v0041 (Baseline) vs v0026 (Best Legal) Setup

The **Best Legal Setup** (v0026) improves the baseline performance while maintaining positive ground clearance. From a vehicle dynamics perspective, configuration v0026 represents an optimal mechanical trade-off:

- **Front Axle** ($K_{front} = 1.3$, $FARB = 0.7$): The front springs are stiffened to support the aerodynamic downforce and limit longitudinal pitch under braking, successfully preventing the front from hitting the ground. To counteract the understeer caused by stiff front springs, the Front Anti-Roll Bar is softened to its minimum.
- **Rear Axle** ($K_{rear} = 1.0$, $RARB = 1.3$): The rear springs are kept at their baseline stiffness, meanwhile, the Rear ARB is stiffened. This shifts the roll stiffness distribution towards the rear axle, promoting a sharper turn-in response and neutralizing the understeering tendency induced by the rigid front springs.

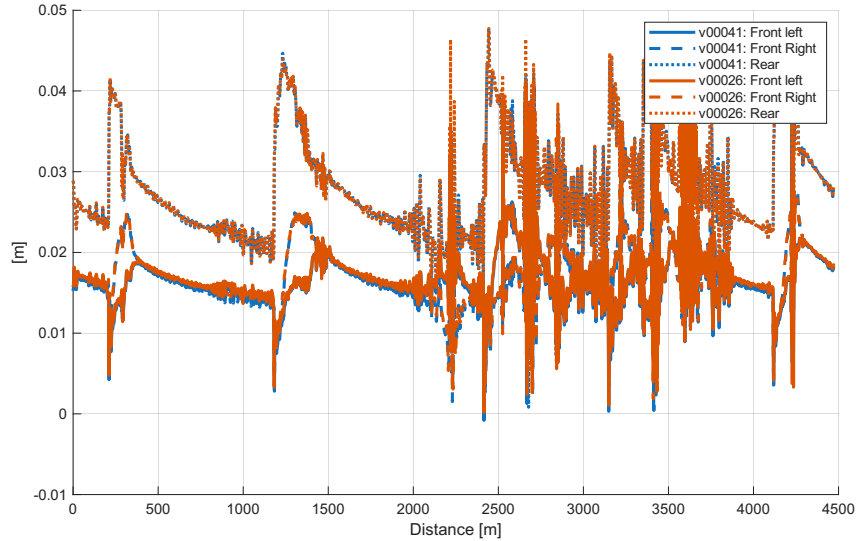


Figure 149: Ride Heights Plots